

Exam Seat no.

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BE Semester-I (June 2023)

Engineering mathematics-I (New) (CC101-N)

Date:16/6/23

Time:3 hour

Marks: 70

- Instruction:** 1) Answer each section in separate Answer sheet.
 2) Use of Scientific calculator is permitted.
 3) All questions are **compulsory**.
 4) Indicate **clearly**, the options you attempt along with its respective question number.
 5) Use the last page of main supplementary for **rough work**.
 6) Use of table for area under the standard normal curve is permitted.

Section I

- Q.1 (i) Evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$ [5]
- (ii) Expand $f(x) = \log x$ in powers of $(x-2)$ by Taylor series method. [5]
- (iii) Find the $J = \frac{\partial(x,y)}{\partial(r,\theta)}$ for $x = r \cos \theta$ and $y = r \sin \theta$. [5]
- OR**
- (iii) If $u = \tan^{-1} \left(\frac{x^2 + y^2}{x - y} \right)$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \sin 2u$ [5]
- Q.2 (i) Find the equations of tangent plane and the normal line to the surface $2x^2 + y^2 + 2z = 3$ at the point $(2, 1, -3)$. [5]
- (ii) Test the convergence of the series $\sum_{n=0}^{\infty} \frac{2^n - 1}{3^n}$. [5]
- OR**
- (i) Trace the curve $r = a(1 + \cos \theta)$. [5]
- (ii) Evaluate $\int_0^{\pi} (1 + \cos \theta)^4 d\theta$. [5]
- Q.3 (i) If $u = f(x-y, y-z, z-x)$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. [5]
- (ii) Evaluate $\int_{-1}^1 \int_0^2 (1 - 6x^2y) dx dy$. [5]
- OR**
- (i) Expand $e^x \cos y$ in powers of x and y up to second degree. [5]
- (ii) Evaluate $\int_0^2 \int_1^3 \int_1^2 xy^2z dz dy dx$. [5]

Section II

Q.4 (i) Find $\frac{\partial z}{\partial x}$ and $\frac{\partial y}{\partial x}$ at (1,-1,2) if $x^2 + y^2 + z^2 = a^2$. [5]

(ii) Evaluate $\lim_{x \rightarrow \frac{\pi}{4}} (1 - \tan x) \sec 2x$ [5]

(iii) Trace the curve $y^2(2a - x) = x^3$. [5]

OR

(iii) Prove that $\int_0^1 \int_1^2 (xy) dy dx = \frac{3}{4}$. [5]

Q.5 (i) Expand $f(x) = x^4 - 11x^3 + 43x^2 - 60x + 14$ in powers of $(x-3)$. [5]

(ii) Show that $\sum_{n=1}^{\infty} \left(\frac{\log n}{1000}\right)^n$ diverges. [5]

OR

(i) Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$ [5]

(ii) If $u = \log(x^2 + y^2)$ then verify $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$. [5]

Q.6 (i) Find the extreme values of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. [5]

(ii) Evaluate $\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy dx$ by changing the order of the integration. [5]

OR

(i) Trace the curve $r^2 = a^2 \cos 2\theta$. [5]

(ii) Find the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$. [5]

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